DA5402 Assignment 8

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Dataset

The Yellow Taxi and Green Taxi datasets were used.

Cleaning Operations:

1. Renaming Pickup and Dropoff times columns
2. Remove all NA values in the columns
3. Format the timestamp columns
4. Remove rows with RatecodeID == 99 (Unknown)
5. Remove rows with unknown or unpaid payments
6. Remove rows with null trip distance
7. Remove inconsistent pickup and dropoff times
8. Remove empty passenger counts

Joining Operations:
join_key: "VendorID"
partition_col: "time_bucket"

Inner Join between Yellow and Green Taxis

Custom Features:
group_col: "PULocationID”

1. Duration in Seconds - Dropoff time - Pickup time
2. Average speed per hour - trip distance / duration (in hr)
3. Revenue tier - total_amount -> High / Medium / Low
4. Zone Average Amount - Mean of total amount
5. Zone Total Passengers - Sum of Passenger count over Pickup location

Systems

2 Laptops were used (Ubuntu and Mac)

Ray

{“framework”: “ray”, “yellow”: 3.03, “green”: 3.03}

Spark

{"framework": "spark", "yellow": 16.13, "green": 16.13}

Conclusion

As we can see, Ray runs faster than Spark for the task